



Armadale Kalamunda Group

Venous Thromboprophylaxis Obstetric Guideline

1. Purpose

Pulmonary embolism (PE) and deep vein thrombosis (DVT) are two components of a single disease called venous thromboembolism (VTE). Thromboembolism is a major cause of maternal mortality. VTE was the leading cause of direct maternal death in Australia 2006–2016. Pregnancy increases the risk of Venous Thromboembolism (VTE) tenfold. Puerperium increases risk 2-5 times. The risk is even greater, 3-16 times, if birth is by caesarean section, particularly emergency caesarean.

Although it is associated with high degrees of morbidity and mortality, VTE during pregnancy and the immediate postnatal period is rare about 1-2/1000 births. Failure to recognise and/or treat personal or pregnancy specific risk factors has been identified as a significant contributing factor to maternal mortality and morbidity arising from VTE in pregnancy.

Risk is higher in third trimester compared with first and second trimester.

Greatest risk is in the weeks immediately after birth. Risk declines to that of general population by about 13 to 18 weeks postpartum and recurrence rate is about 2–11%

The purpose of this guideline is to recognise the need for thromboprophylaxis in women at high risk of thromboembolism in pregnancy and the postpartum period and screen all pregnant women at risk.

2. Applicability

This guideline applies to midwifery, medical and anaesthetics staff within the obstetric unit at Armadale Health Service.

3. Guidelines

3.1 Antenatal risk assessment

There is 4 to 5-fold increased risk of developing a VTE in pregnancy due to:

- Hypercoagulability
- Increased venous stasis
- Decreased venous outflow
- Compression of the inferior vena cava and pelvic veins by the gravid uterus
- Decreased mobility
- Altered levels of coagulation factors responsible for haemostasis

Antenatal risk assessment needs to be completed at booking for all women AKMR23.1 (see flow chart [appendix 1](#))

3.2 Prophylaxis

3.2.1 Non-pharmacological methods:

- Ted stockings, pneumatic compression (Flowtron), early mobilisation, hydration and surveillance.

3.2.2 Pharmacological methods:

- Low molecular weight heparin (LMWH)
- Unfractionated heparin

3.3 Prophylaxis of VTE with Enoxaparin (LMWH)

Where level of risk indicates the need for chemical thromboprophylaxis, the following is a guideline for treatment:

- Collaborative management may be required involving a haematologist and/or anaesthetist. Liaise with Fiona Stanley Hospital Haematologist team.
- Consider anaesthetic referral from 32 weeks
- Consider planned birth if indicated.
- Plan intrapartum care.
- Consider precautions form neuraxial blockade. Discontinue anticoagulant 24 hours prior to induction of Labour/ Caesarean section. Send blood for cross match and consult Specialist Obstetrician/ anaesthetist for discussion regarding possible tertiary referral if spontaneous labour.

3.4 Antenatal and postnatal prophylaxis dosage

Enoxaparin (Clexane®) is prescribed subcutaneously for prophylaxis in antenatal patients with risk factors and for women requiring prolonged immobilisation antenatally. ([see appendix 2](#))

Neither heparin (unfractionated or LMWH) or warfarin is contraindicated in breastfeeding. There are few published data on whether LMWH's are secreted in breast milk, although extensive experience of enoxaparin in the puerperium reports no problems during breastfeeding and other heparins are known not to cross the breast.

4. Thrombophilia associated VTE

4.1 Types

- Inherited - Antithrombin deficiency, Protein S or C deficiency, Activated protein C resistance/ factor V Leiden, prothrombin gene mutation
- Acquired- antiphospholipid antibodies/ lupus anticoagulant

4.2 Management of Thrombophilia associate VTE in pregnancy: ([see appendix 3](#))

5. Pre and post caesarean birth

All women, especially caesarean section women, are screened once postnatally using the AKMR 23.3 Postnatal Venous Thromboembolism (VTE) Risk Assessment Tool.

All non-elective caesarean section women will receive VTE prophylaxis and elective caesarean women will be assessed for risk. All women with previous VTE should receive post-natal prophylaxis.

Enoxaparin (Clexane®) 40mg subcutaneously is prescribed for prophylaxis postnatally for all non-elective caesarean section women and women with risk factors for VTE.

Insufficient data is available to recommend increasing the prophylactic dose of LMWH for women with weight >90 kg; (group consensus). Consider high prophylactic dose of 40 IU BD for morbidly obese women with risk factors after discussion with Specialist Obstetrician and anaesthetist.

- Epidural catheter should not be removed until 12 hours after a prophylactic dose of anticoagulant and 24 hours post therapeutic dose.
- First dose to be administered no sooner than 4 hours post-delivery and no later than 6 hours post-operatively.
- Aim to prescribe LMWH dose at either 08:00hrs or 20:00 hours.
- If time between surgical procedure and 08:00hrs or 20:00hrs exceed 6 hours consider giving bridging dose of unfractionated heparin
- Treatment is to be continued for at least 7days in moderate risk category.
- Treatment is to be continued for a minimum of 6 weeks in high-risk category or longer as physician recommends.
- Neither heparin (unfractionated or LMWH) or warfarin is contraindicated in breastfeeding. There are few published data on whether LMWH's are secreted in breast milk, although extensive experience of enoxaparin in the puerperium reports no problems during breastfeeding and other heparins are known not to cross the breast.

6. Diagnosing Deep Vein Thrombosis (DVT) / Pulmonary Embolism (PE)

- Do not use D-dimer testing in pregnancy.
- Doppler ultrasonography or compression ultrasonography to confirm DVT.
- Ventilation perfusion (VQ) scan or Computerised Pulmonary Angiography (CTPA) to confirm Pulmonary embolism.

6.1 Treatment DVT/PE

- Confirmed pulmonary embolism or DVT above the knee: therapeutic dosage for minimum 6 months and prophylaxis for rest of pregnancy until 6 weeks postpartum.
- Confirmed DVT in calf: therapeutic dosage for minimum 3 months and prophylactic dose till 6 weeks postpartum, provided thrombus resolution and no post thrombosis syndrome.
- Enoxaparin 1mg/kg twice/day or 1.5mg/kg daily for therapeutic anticoagulation dose

Note: LMWH should be avoided, discontinued, or postponed in women who are at risk of bleeding after careful consideration of the balance of risks of bleeding and clotting.

7. Risk factors for bleeding:

- Women with active antenatal or postpartum bleeding.
- Primary postpartum haemorrhage (PPH) greater than 1L
- Women considered at increased risk of major haemorrhage (such as placenta previa)
- Women with a bleeding diathesis, such as von Willebrand's disease, haemophilia or acquired coagulopathy.
- Women with thrombocytopenia (platelet count less than 75×10^9)

- Acute stroke in the last 4 weeks (ischaemic or haemorrhagic)
- Severe renal disease (glomerular filtration rate less than 30 ml/minute/1.73 m²)
- Severe liver disease (prothrombin time above normal range or known varices)
- Uncontrolled hypertension (blood pressure greater than 200 mmHg systolic or greater than 120 mmHg diastolic).
- Haemorrhage prone ulcerative colitis.
- Chronic, clinically significant, and measurable bleeding over 48 hours
- Endocarditis
- IMI administration of medication

Further advice on the decision and the management of both VTE risk factors and bleeding risk factors may be sought from a haematologist with experience in the management of thrombosis and bleeding disorders in pregnancy.

There is no indication for the use of Flowtrons as well as TEDS with Clexane

8. Legislative context

The following documents are required to give effect to this policy:

- Health Department of Western Australia (HDWA) [High Risk Medication Policy MP 0131/20](#)
- HDWA [Clinical Handover Policy MP 0095](#)
- HDWA [Recognising and Responding to Acute Deterioration Policy MP 0086/18](#)
- East Metropolitan Health Service (EMHS) [Consent Policy EMHS: 5](#)

9. Supporting information

The following documents support the implementation of this policy:

- Antenatal Venous Thromboembolism (VTE) Risk Assessment Tool AKMR 23.1
- Pre-Operative Autar DVT Risk Assessment Scale AKMR 23
- Postnatal Venous Thromboembolism (VTE) Risk Assessment Tool AKMR 23.2
- [Antimicrobial Stewardship Policy AKG-POL-0039-18](#)
- [Surgical Prophylaxis Guideline AKG-GUI-0548-20](#)
- [Venous Thromboembolism Prevention Guideline – Adult AKG-GUI-0524-20](#)
- [Non-Elective Caesarean Section Guideline OBS-GUI-0302-10](#)

10. Compliance, monitoring and evaluation

Compliance against this guideline will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Department specific findings are to be reported at the departmental meeting for review and identification of any opportunities for improvement.

11. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.15 Antimicrobial stewardship
- Standard 4 – Action 4.3 Partnering with consumers.
- Standard 4 – Action 4.15 High-risk medicines
- Standard 5 – Action 5.11 Clinical assessment
- Standard 6 – Action 6.5 Correct identification and procedure matching
- Standard 6 – Action 6.7 Clinical handover
- Standard 8 – Action 8.4 Recognising acute deterioration.

12. References

1. RCOG Green Top Guideline No: 37a, reducing the risk of venous Thromboembolism during Pregnancy and Puerperium: April 2015
2. RCOG Green Top Guideline No: 37b Thrombosis and Embolism during Pregnancy and the Puerperium, April 2015
3. KEMH guidelines: VTE prevention and management 2021
4. Queensland Clinical Guidelines: Venous thromboembolism (VTE) prophylaxis in pregnancy and the puerperium 2020

13. Glossary

The following definitions are relevant to this policy.

Term	Definition
Venous Thromboembolism	A condition where a blood clot forms most often in the deep veins of the leg, groin or arm (known as deep vein thrombosis, DVT) and travels in the circulation, lodging in the lungs (known as pulmonary embolism, PE).

14. Document owner

Director Obstetrics & Gynaecology

Enquiries relating to this guideline may be directed to:

Title: Head of Department
 Department: Obstetrics & Gynaecology
 Email: Sangeeta.MallaBhat@health.wa.gov.au

15. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4	15/04/2025	15/04/2028	Appendices updated

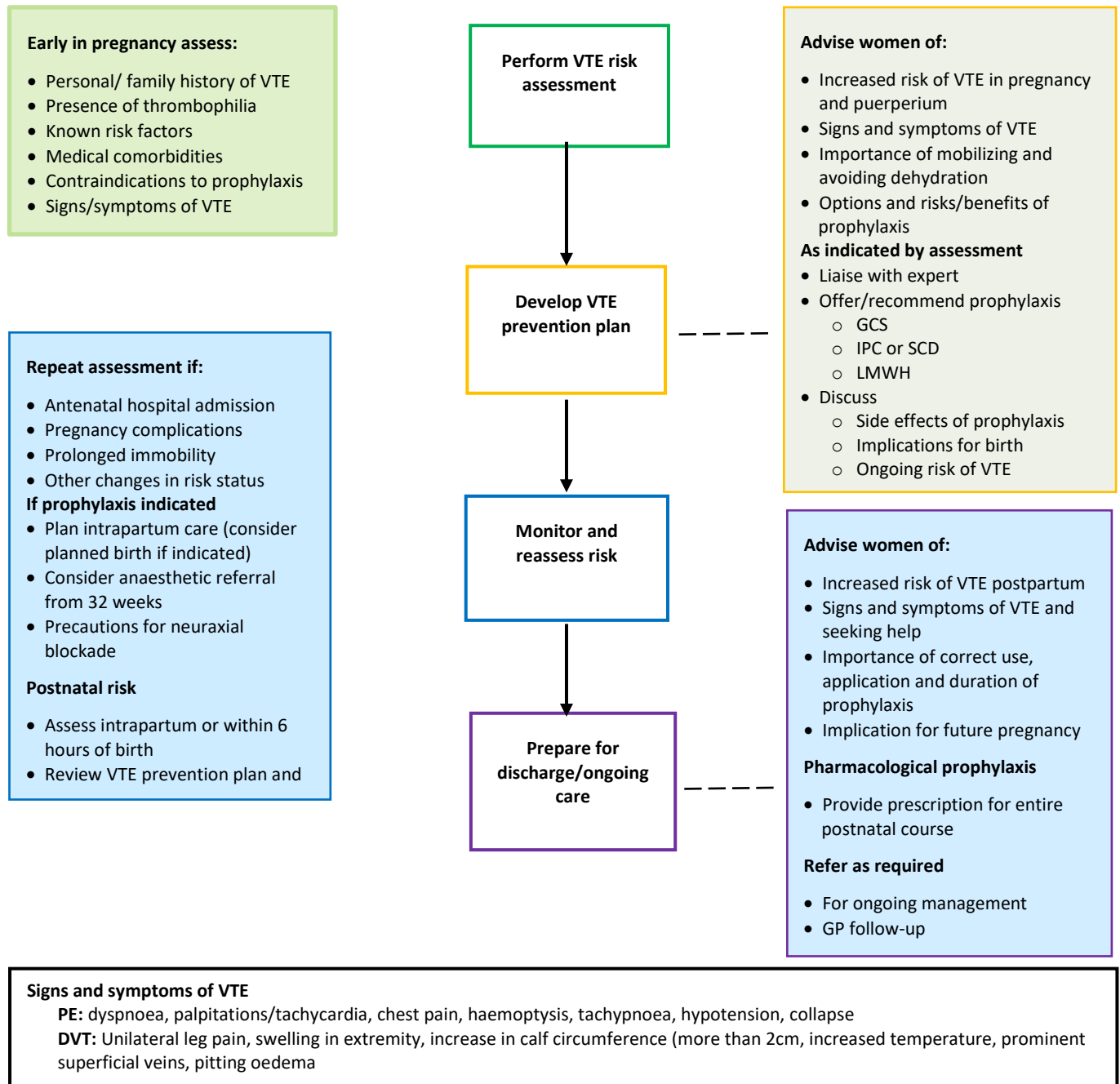
16. Authorisation

This guideline has been authorised and issued by the Director of Clinical Services.

Authorised by	Dr Tania Strickland – Director of Clinical Services
Authorisation date	09/09/2021
Published date	14/09/2021

Appendix 1: VTE assessment for pregnant and postpartum women

Assess women on an individual basis.



DVT: deep vein thrombosis, **GCS:** graduated compression stockings, **GP:** general practitioner, **IPC:** intermittent pneumatic compressions, **LMWH:** low molecular weight heparin, **PE:** pulmonary embolism, **SCD:** sequential compression device, **VTE:** venous thromboembolism

Appendix 2: Antenatal Postnatal thromboprophylaxis according to risk

High risk	1	ANY ONE OF ☐ Pre-pregnancy therapeutic anticoagulation (any reason) ☐ Any previous VTE plus high-risk thrombophilia** ☐ Recurrent unprovoked VTE (2 or more) ☐ VTE in current pregnancy (seek expert advice)	Therapeutic anticoagulation • Continue/commence antenatal • Continue 6 weeks postpartum <i>*High prophylactic dose may be appropriate</i>
	2	ANY ONE OF ☐ Any single previous VTE not provoked by surgery ☐ Recurrent provoked VTE (2 or more) ☐ Active autoimmune of inflammatory disorder ☐ Medical co-morbidity: (e.g., cancer, nephrotic syndrome, heart failure, sickle cell, type 1 diabetes with nephropathy)	LMWH standard prophylaxis • From first trimester • Continue 6 weeks postpartum
	3	IF THROMBOPHILIA ☐ High or low risk thrombophilia *(no personal history VTE)	Refer to Flowchart:
	4	ANY ONE OF ☐ Antenatal hospital admission ☐ Ovarian hyperstimulation syndrome (first trimester only) ☐ Any surgery (pregnancy or postpartum) ☐ Severe hyperemesis or dehydration requiring IV fluid	LMWH Standard prophylaxis • While in hospital or until resolves

All risk	5	SELECT ALL THAT APPLY at every assessment (antenatal or postnatal) <table border="1"> <thead> <tr> <th></th> <th>Risk Score</th> </tr> </thead> <tbody> <tr><td>☐ Family history (1st degree relative) of unprovoked or estrogen provoked VTE</td><td>1</td></tr> <tr><td>☐ Single VTE provoked by surgery</td><td>3</td></tr> <tr><td>☐ Age >35 years</td><td>1</td></tr> <tr><td>☐ Parity ≥3</td><td>1</td></tr> <tr><td>☐ Smoking (any amount)</td><td>1</td></tr> <tr><td>☐ Gross varicose veins</td><td>1</td></tr> <tr><td>☐ Current BMI 30-39 kg/m²</td><td>1</td></tr> <tr><td>☐ Current BMI ≥40 kg/m²</td><td>2</td></tr> <tr><td>☐ IVF/ART</td><td>1</td></tr> <tr><td>☐ Multiple pregnancy</td><td>1</td></tr> <tr><td>☐ Pre-eclampsia in current pregnancy</td><td>1</td></tr> <tr><td>☐ Immobility</td><td>1</td></tr> <tr><td>☐ Current systemic infection</td><td>1</td></tr> <tr><td>☐ Pre-existing diabetes</td><td>1</td></tr> <tr><td colspan="2"> </td></tr> <tr><td>☐ Caesarean section in labour</td><td>3</td></tr> <tr><td>☐ Elective caesarean section</td><td>1</td></tr> <tr><td>☐ Prolonged labour > 24 hours</td><td>1</td></tr> <tr><td>☐ Operative vaginal birth</td><td>1</td></tr> <tr><td>☐ Preterm birth (<37+0 weeks)</td><td>1</td></tr> <tr><td>☐ PPH > 1L or transfusion</td><td>1</td></tr> <tr><td>☐ Stillbirth in current pregnancy</td><td>1</td></tr> <tr><td>☐ Caesarean hysterectomy</td><td>3</td></tr> </tbody> </table>		Risk Score	☐ Family history (1 st degree relative) of unprovoked or estrogen provoked VTE	1	☐ Single VTE provoked by surgery	3	☐ Age >35 years	1	☐ Parity ≥3	1	☐ Smoking (any amount)	1	☐ Gross varicose veins	1	☐ Current BMI 30-39 kg/m ²	1	☐ Current BMI ≥40 kg/m ²	2	☐ IVF/ART	1	☐ Multiple pregnancy	1	☐ Pre-eclampsia in current pregnancy	1	☐ Immobility	1	☐ Current systemic infection	1	☐ Pre-existing diabetes	1			☐ Caesarean section in labour	3	☐ Elective caesarean section	1	☐ Prolonged labour > 24 hours	1	☐ Operative vaginal birth	1	☐ Preterm birth (<37+0 weeks)	1	☐ PPH > 1L or transfusion	1	☐ Stillbirth in current pregnancy	1	☐ Caesarean hysterectomy	3	Antenatal risk score <table border="1"> <tr> <td>All</td> <td>Mobilise early, avoid dehydration</td> </tr> <tr> <td>3</td> <td> LMWH standard prophylaxis • From 28 weeks </td> </tr> <tr> <td>≥4</td> <td> LMWH standard prophylaxis • From time of assessment </td> </tr> </table>	All	Mobilise early, avoid dehydration	3	LMWH standard prophylaxis • From 28 weeks	≥4	LMWH standard prophylaxis • From time of assessment
		Risk Score																																																							
☐ Family history (1 st degree relative) of unprovoked or estrogen provoked VTE	1																																																								
☐ Single VTE provoked by surgery	3																																																								
☐ Age >35 years	1																																																								
☐ Parity ≥3	1																																																								
☐ Smoking (any amount)	1																																																								
☐ Gross varicose veins	1																																																								
☐ Current BMI 30-39 kg/m ²	1																																																								
☐ Current BMI ≥40 kg/m ²	2																																																								
☐ IVF/ART	1																																																								
☐ Multiple pregnancy	1																																																								
☐ Pre-eclampsia in current pregnancy	1																																																								
☐ Immobility	1																																																								
☐ Current systemic infection	1																																																								
☐ Pre-existing diabetes	1																																																								
☐ Caesarean section in labour	3																																																								
☐ Elective caesarean section	1																																																								
☐ Prolonged labour > 24 hours	1																																																								
☐ Operative vaginal birth	1																																																								
☐ Preterm birth (<37+0 weeks)	1																																																								
☐ PPH > 1L or transfusion	1																																																								
☐ Stillbirth in current pregnancy	1																																																								
☐ Caesarean hysterectomy	3																																																								
All	Mobilise early, avoid dehydration																																																								
3	LMWH standard prophylaxis • From 28 weeks																																																								
≥4	LMWH standard prophylaxis • From time of assessment																																																								
	<table border="1"> <tr> <td colspan="2"> Postnatal risk score = antenatal + postnatal score </td> </tr> <tr> <td>All</td> <td>Mobilise early, avoid dehydration</td> </tr> <tr> <td>2</td> <td> LMWH standard prophylaxis • Until discharge </td> </tr> <tr> <td>≥3</td> <td> LMWH standard prophylaxis • 7 days (or longer if ongoing risk) </td> </tr> <tr> <td colspan="2"> All caesarean sections • Recommend IPC or SCD until next day </td> </tr> </table>	Postnatal risk score = antenatal + postnatal score		All	Mobilise early, avoid dehydration	2	LMWH standard prophylaxis • Until discharge	≥3	LMWH standard prophylaxis • 7 days (or longer if ongoing risk)	All caesarean sections • Recommend IPC or SCD until next day																																															
Postnatal risk score = antenatal + postnatal score																																																									
All	Mobilise early, avoid dehydration																																																								
2	LMWH standard prophylaxis • Until discharge																																																								
≥3	LMWH standard prophylaxis • 7 days (or longer if ongoing risk)																																																								
All caesarean sections • Recommend IPC or SCD until next day																																																									

Sum all risk scores	
----------------------------	--

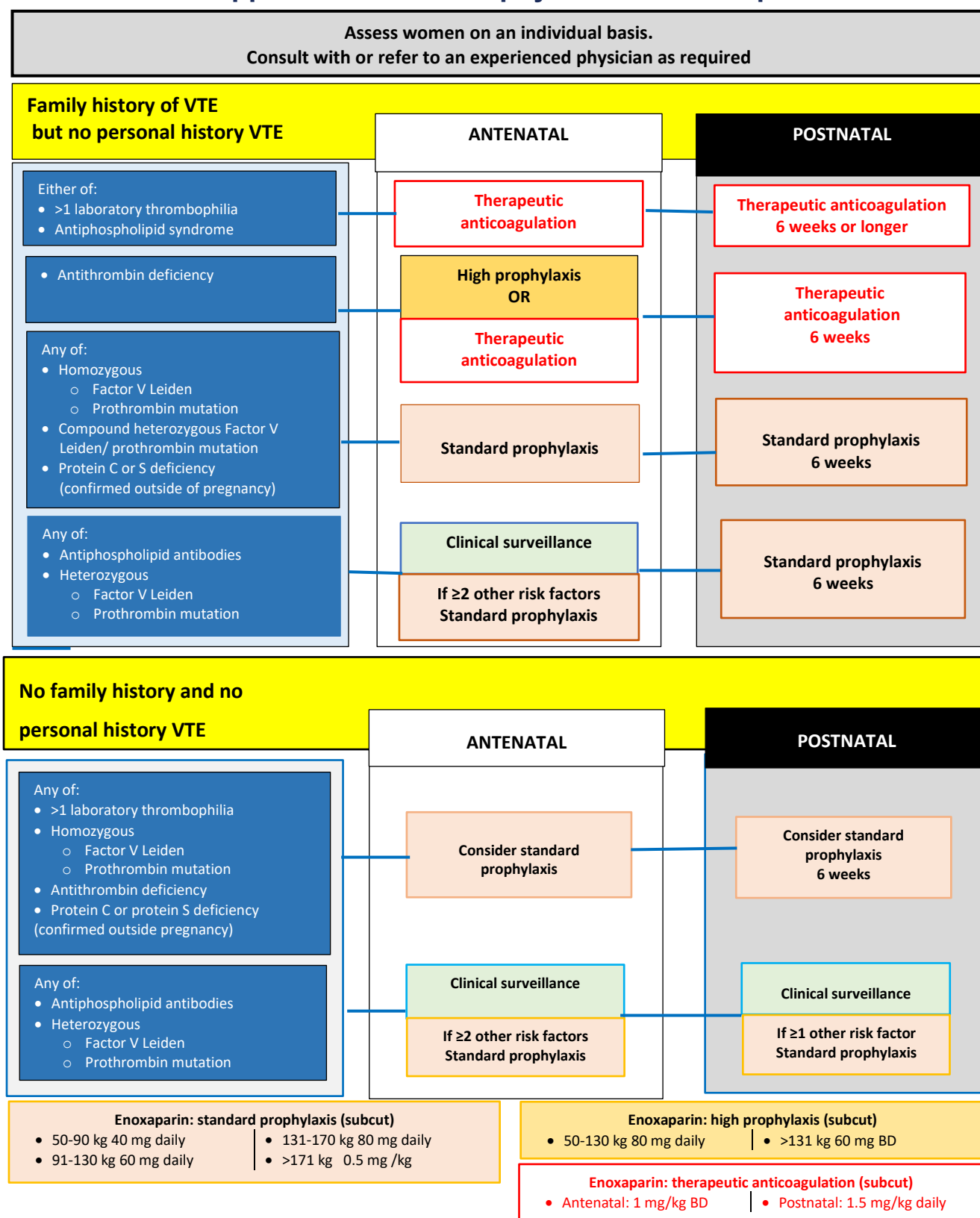
***High risk thrombophilia:** >1 laboratory thrombophilia, APS, antithrombin deficiency, protein C deficiency, protein S deficiency, homozygous FVL, homozygous prothrombin mutation, compound heterozygous FVL/prothrombin mutation

Low risk thrombophilia: heterozygous FVL, heterozygous prothrombin mutation, antiphospholipid antibodies

GCS/TED stockings Consider for postnatal women until fully mobile Recommend if receiving LMWH	Enoxaparin: standard prophylaxis (subcut) <table border="1"> <tr> <td>• 50-90 kg 40 mg daily</td> <td>• 131-170 kg 80 mg daily</td> </tr> <tr> <td>• 91-130 kg 60 mg daily</td> <td>• >171 kg 0.5 mg/kg</td> </tr> </table>	• 50-90 kg 40 mg daily	• 131-170 kg 80 mg daily	• 91-130 kg 60 mg daily	• >171 kg 0.5 mg/kg
• 50-90 kg 40 mg daily	• 131-170 kg 80 mg daily				
• 91-130 kg 60 mg daily	• >171 kg 0.5 mg/kg				

APS: antiphospholipid syndrome, **ART:** artificial reproductive technology, **BMI:** body mass index, **FVL:** factor V Leiden, **GCS:** graduated compression stockings, **IPC:** intermittent pneumatic compressions, **IVF:** in-vitro fertilization, **LMWH:** low molecular weight heparin, **PE:** pulmonary embolism, **PPH:** primary postpartum haemorrhage, **SCD:** sequential compression device, **SLE:** systemic lupus erythematosus, **TEDS:** thromboembolic deterrent stockings, **VTE:** venous

Appendix 3: Thromboprophylaxis if thrombophilia



High risk thrombophilia: > 1 laboratory thrombophilia, APS, antithrombin deficiency, protein C deficiency, protein S deficiency, homozygous FVL, homozygous prothrombin mutation, compound heterozygous FVL /prothrombin mutation

Low risk thrombophilia: heterozygous FVL, heterozygous prothrombin mutation, antiphospholipid antibodies

APS: antiphospholipid syndrome, FVL: Factor V Leiden, BD: twice daily